Power law attention biases for molecular transformers

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Why predict molecular properties?

Properties like energies, forces, and orbital levels drive drug discovery, catalysis, and materials design.

The gold standard—density functional theory (DFT)—is accurate but expensive: minutes to hours per molecule, scaling poorly with system size.

Goal: train a neural network on DFT-labeled data, then run inference in milliseconds.

What does the data look like?

A molecule is a 3D point cloud:

- Atomic numbers $z_i \in \mathbb{N}$
- 3D coordinates $\mathbf{r}_i \in \mathbb{R}^3$

Scalar targets (e.g. total energy) must be invariant under:

- Translations and rotations of the whole molecule
- Permutations of atom indices

No canonical ordering, no grid, no absolute coordinates—just relative geometry.

The molecular ML landscape

Equivariant graph neural networks dominate molecular property prediction[1][2].

Transformers, despite ruling vision and language, are rarely competitive here. Why the asymmetry?

Hypothesis: transformers lack good *inductive biases* for many-body geometry.

[1] Li, Zhang, Shen, et al. 2025.

[2] Jacobs et al. 2025.

The problem with dot-product attention

The root of the problem lies in the attention mechanism:

$$A_{ij} = \mathrm{softmax}_j\left(\frac{\mathbf{q}_i \mathbf{k}_j^T}{\sqrt{d_k}}\right)$$

This form implies the token embedding must encode *both* the token's identity *and* its relationships to other tokens.

- → Fine for language/images, absolute token indices $\implies$ positional encodings
- → Bad for point clouds: no natural indices and structure is continuous.

Attention biases

Consider adding a pairwise term b_{ij} inside the softmax:

$$A_{ij} = \mathrm{softmax}_j\left(\frac{\mathbf{q}_i \mathbf{k}_j^T}{\sqrt{d_k}} + b_{ij}\right)$$

- Permutation invariance for free.
- Naturally *pairwise*, just like all physical interactions.
- **Easy to construct given a notion of distance between tokens.**

Power law attention biases

Consider the $E(3)$-equivariant bias $b_{ij} = p \log ||\mathbf{r}_i - \mathbf{r}_j||$.

Through the softmax:

$$A_{ij} = \text{softmax}_j \left[\frac{\mathbf{q}_i \mathbf{k}_j^T}{\sqrt{d_k}} + p \log ||\mathbf{r}_i - \mathbf{r}_j|| \right]$$
$$\propto ||\mathbf{r}_i - \mathbf{r}_j||^p \exp \left[\frac{\mathbf{q}_i \mathbf{k}_j^T}{\sqrt{d_k}} \right]$$

The bias *weights* attention by a power of the interatomic distance.

Analogous to Coulomb's law: interaction strength is set by both charges (embeddings) and distance.

Design choices for p

The remaining degrees of freedom lie in p. Several options:

- Fix $p = -1$ (echoes potential-energy fall-off).
- Learn $p \in \mathbb{R}$.
- Learn $p \in \mathbb{R}_-$ via reparameterization $p = -e^{\theta}$.

Edge case: singularity when $i = j$. Resolve by masking out the diagonal.

Setup

Standardized transformer architecture across all baselines:

Embedding dimension	128
Number of layers	8
Number of attention heads	8
Dropout	0.1
Number of epochs	128
Batch size	64
Learning rate	0.0001
Weight Decay	0.00001

Roughly 1.5M parameters per model, trains in 1 hour on 1 V100 GPU.

Benchmarks

We train and evaluate on two DFT datasets:

- **QM9**: 134k molecules (5M tokens) labeled with orbital energies HOMO, LUMO and state energies U, H, G.
- **SPICE**: 1.1M molecules (50M tokens) labeled with formation energy U_F.

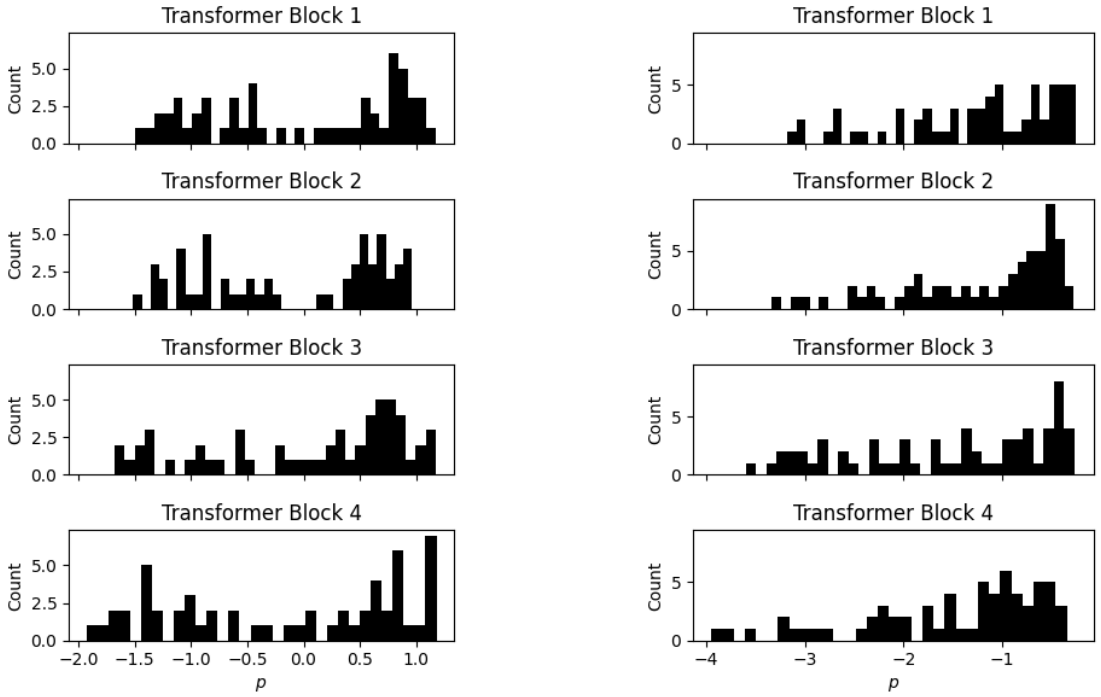
Results

	QM9					SPICE
Structural bias	HOMO	LUMO	U	H	G	U_F
Transformer	0.34	0.74	24	24	24	10
Graph Attention	0.12	0.14	29	29	30	-
RWPE	0.14	0.20	42	42	42	-
Attention bias						
Gaussian kernel	0.08	0.10	21	21	21	-
Power law $p = -1$	0.11	0.13	26	26	26	7
Power law $p \in \mathbb{R}$	0.12	0.14	25	25	25	5
Power law $p \in \mathbb{R}_-$	0.11	0.12	21	21	21	5

Table: Test ensemble MAE (eV).

Attention biases beat positional encodings and graph attention, while matching Gaussian kernels at a fraction of the parameters.

Learned exponents



(a) $p \in \mathbb{R}$ (b) $p \in \mathbb{R}^-$

Figure: Histograms of learned exponents over 7-model ensembles.

Exponents don't tend to any specific power law and when unconstrained even drift positive (unphysical).

Ablating dot-product attention

How much modeling power do power law attention biases have? Test by ablating the dot-product:

$$A_{ij} = \mathrm{softmax}_j\left(\frac{\mathbf{q}_i \mathbf{k}_j^T}{\sqrt{d_k}} + b_{ij}\right) \quad \rightarrow \quad A_{ij} = \mathrm{softmax}_j(b_{ij})$$

This form is basically weighted message passing on a complete graph, and structure is fully decoupled from token content.

Bonus: no dot products $\implies$ cheaper to evaluate.

Effect of ablation

	HOMO	LUMO	U	H	G
Transformer	0.34	0.74	24	24	24
Attention bias					
Gaussian kernel	0.1 (+8%)	0.1 (+7%)	20 (-5%)	20 (-6%)	20 (-7%)
Power law $p = -1$	0.1 (+14%)	0.2 (+19%)	27 (+3%)	27 (+2%)	27 (+4%)
Power law $p \in \mathbb{R}$	0.1 (+9%)	0.1 (+11%)	26 (+3%)	25 (+2%)	25 (+3%)
Power law $p \in \mathbb{R}_-$	0.1 (+11%)	0.1 (+12%)	25 (+9%)	25 (+8%)	25 (+9%)

Table: Ensemble MAE (eV); percent change from unablated in parentheses.

- Loss increases modestly—and ablated models still beat the vanilla transformer.
- Gaussian kernel biases actually *improve* on U, H, G when the dot-product is removed.

Takeaways

Power law attention biases are:

- Simple and physically motivated.
- More effective than most other inductive biases.
- Only matched by Gaussian kernels, which are much more expensive.
- Highly effective on their own, as shown by ablation.

Caveat: Uninterpretable learned exponents $\implies$ physicality of the mechanism is dubious.

Questions/Limitations

- How do these results change with scale?
- How do these results change out-of-distribution?
- How accurate and useful are first derivatives (forces)?
- Are predictions stable in dynamical settings?
- How do initializations affect learned p?
- Are models inhibited by gradient flow issues in p?
- Can message passing layers cheapen large models?

End